Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 1: Nutrition in Plants

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	1 — Nutrition in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: assuming plants stop respiring during the day or stop living at night, confusing the chlorophyll-removal (alcohol) and softening (boiling-in-water) starch-test steps, mixing up xylem versus phloem direction, calling insectivorous plants non-photosynthetic, and treating parasites as saprotrophs or symbionts. Do not write on this paper — mark answers on your answer sheet.

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1. A scientist measures the gases leaving a single bean leaf over 24 hours. Net oxygen output is positive between 7 a.m. and 6 p.m. and negative (net oxygen is taken IN) the rest of the time. At exactly two instants the net oxygen exchange is zero. What is happening at those two zero-crossing instants?

(A) The leaf has temporarily stopped both photosynthesis and respiration.
(B) The rate of photosynthesis exactly equals the rate of respiration, so oxygen made equals oxygen used.
(C) The leaf is only respiring and using up all its own oxygen internally.
(D) Photosynthesis has completely stopped while respiration continues at full speed.

2. Two leaf discs are cut from the same plant. Disc P is boiled in water for 2 minutes, then boiled in alcohol, then dipped in iodine. Disc Q is dipped straight into iodine with no boiling. Disc P shows a clear blue-black colour; disc Q shows almost nothing even though both discs contained starch. The best explanation for Q's poor result is:

(A) Iodine could not penetrate the living, green, waxy disc, so the starch present went undetected.
(B) Boiling in water created the starch that disc P shows.

(C) Disc Q never contained any starch in the first place.

(D) Alcohol adds starch to a leaf, which is why disc P reacted.

3. A student claims: 'Since plants release oxygen, a bedroom full of houseplants is dangerous to sleep in at night because the plants will steal the oxygen.' Which single fact most directly supports the WORRY (that oxygen could drop) while still being scientifically correct?

(A) Plants photosynthesise faster at night than in the day.

(B) Plants have no need for oxygen at any time.

(C) At night plants carry out only respiration, taking in oxygen and giving out carbon dioxide.

(D) Chlorophyll absorbs oxygen directly from the air in darkness.

4. A pond contains green algae, tadpoles that eat algae, and small fish that eat tadpoles. If a disease suddenly kills almost all the green algae, the MOST immediate and direct consequence is:

(A) The fish population rises because there are fewer algae competing with them.

(B) Tadpoles lose their food supply and their numbers fall.

(C) The tadpoles switch to making their own food from sunlight.

(D) The fish die first, before any change reaches the tadpoles.

5. A potted geranium is destarched by keeping it in the dark for two days. One healthy leaf is then fitted into a flask whose air has had its carbon dioxide removed by a small dish of potassium hydroxide; another leaf on the same plant is left in ordinary air. The whole plant is put in bright light for six hours, then both leaves are tested with iodine. The expected results are:

(A) Both leaves turn blue-black, because light alone makes starch.

(B) The ordinary-air leaf turns blue-black; the carbon-dioxide-free leaf does not.

(C) Neither leaf turns blue-black, because destarching removed their ability to make starch.

(D) Only the carbon-dioxide-free leaf turns blue-black, because removing a gas speeds up food-making.

6. Cuscuta (amarbel) and a mushroom are both heterotrophs that lack chlorophyll, yet they are placed in different nutrition categories. The key difference is that:

(A) Cuscuta makes some of its own food, while the mushroom makes none.

(B) The mushroom takes food from a living host, while Cuscuta feeds on dead matter.

(C) Cuscuta traps insects, while the mushroom fixes nitrogen.

(D) Cuscuta draws food from a living host, while the mushroom feeds on dead, decaying matter.

7. In a leaf, water travels upward in one type of tube and the food made in the leaf travels outward in another. A ringing experiment removes a strip of the OUTER tissue (phloem) all around a young branch but leaves the inner wood (xylem) intact. Over the following weeks the branch is most likely to:

(A) Wilt immediately because water can no longer reach the leaves.

(B) Grow faster than ever, because removing phloem boosts photosynthesis.

(C) Stop making food at once because the leaves lose their water supply.

(D) Keep receiving water but swell above the ring as sugars pile up, since food cannot move down.

8. A farmer rotates her field as follows: year 1 wheat, year 2 wheat, year 3 a legume (gram) which she ploughs back into the soil, year 4 wheat. The soil nitrogen, measured each autumn, is HIGHEST after which crop, and why?

(A) After year 1, because the very first wheat crop adds nitrogen as it grows.

(B) After year 3, because Rhizobium in the gram's root nodules fixed atmospheric nitrogen into the soil.

(C) After year 4, because wheat slowly returns nitrogen to the soil over time.

(D) After year 2, because growing the same crop twice doubles the soil's nitrogen.

9. A variegated leaf (green at the edges, pale white in the middle) is left in sunlight all day, then half of it is additionally covered with foil during a SECOND sunny day before the starch test. Considering both colour and light, which region will turn blue-black?

(A) The whole green area, even the foil-covered part.

(B) The part that is both green AND was left uncovered (exposed to light).

(C) The white centre, because pale tissue stores the most starch.

(D) The foil-covered region, because shading concentrates starch.

10. A lichen is described as a partnership where a fungus provides water, minerals and shelter while an alga provides food. A student says this makes the lichen behave 'just like a parasite living off the alga.' Why is the student wrong?

(A) Because the alga harms the fungus, making it the parasite instead.

(B) Because lichens contain no living organisms, only minerals.

(C) Because the fungus makes the food and the alga only stores it.

(D) Both partners benefit, so it is symbiosis, not parasitism where one is harmed.

11. Four sealed transparent jars are set up in bright light, each holding one organism for a day, then the carbon dioxide level inside is measured. Jar 1: a green fern. Jar 2: a mushroom. Jar 3: a Cuscuta vine (no host). Jar 4: a snail. In which jar is the carbon dioxide level most likely to have FALLEN?

(A) Jar 2, the mushroom.
(B) Jar 1, the green fern.

(C) Jar 3, the Cuscuta vine.
(D) Jar 4, the snail.

12. A teacher writes the photosynthesis word equation and then deliberately introduces ONE error: 'carbon dioxide + oxygen → (sunlight, chlorophyll) → glucose + water.' Which correction fixes the equation?

(A) Replace 'chlorophyll' with 'soil minerals'.

(B) Replace 'glucose' on the right with 'starch'.

(C) Replace 'oxygen' on the left with 'water', and 'water' on the right with 'oxygen'.

(D) Replace 'sunlight' with 'darkness'.

13. Stomata on most leaves tend to close on a very hot, dry, windy afternoon. A direct consequence of these closed stomata during that afternoon is that photosynthesis slows because:

(A) Sunlight can no longer pass through the closed pores.

(B) Chlorophyll is destroyed whenever stomata close.

(C) Carbon dioxide can no longer enter the leaf in sufficient amounts.

(D) Water stops being absorbed by the roots when stomata close.

14. A pitcher plant grows happily in a bog and clearly has bright green leaves, yet it still traps and digests insects. A correct conclusion combining BOTH facts is:

(A) It photosynthesises for its energy-rich food but traps insects to obtain nitrogen the poor soil lacks.

(B) The green colour is decorative; it gets all its food from insects.

(C) It is a parasite, since it feeds on the living insects it traps.

(D) It is a saprotroph that has evolved green leaves by accident.

15. An aquatic plant (Hydrilla) is placed in a beaker of water in bright light, and bubbles of gas rise steadily from it. The same plant in the dark produces no such bubbles. The gas in the bubbles is mostly:

(A) Oxygen released during photosynthesis.

(B) Carbon dioxide released during photosynthesis.

(C) Water vapour escaping from the leaves.(D) Nitrogen taken in from the air.

16. A student lists organisms as autotrophs: (i) mango tree, (ii) green alga, (iii) cactus with green stems, (iv) Cuscuta vine, (v) cyanobacteria-like blue-green organisms that photosynthesise. Exactly one entry is WRONG. Which one, and why?

(A) (iii) the cactus, because stems can never photosynthesise.

(B) (ii) the green alga, because algae are too small to be autotrophs.

(C) (i) the mango tree, because large trees buy energy from the soil.

(D) (iv) Cuscuta, because it lacks chlorophyll and takes ready-made food from a host.

17. Two identical destarched plants are kept in light all day. Plant A's pot soil is normal; plant B's soil has been sterilised to kill nitrogen-fixing bacteria and given no fertiliser. After several WEEKS, comparing a fresh leaf from each by the iodine test on a single sunny day:

(A) Only plant A turns blue-black, because nitrogen is a raw material for the starch test.

(B) Both leaves can still turn blue-black, because the day's photosynthesis needs CO_2, water, light and chlorophyll — not soil nitrogen.

(C) Neither turns blue-black, because without nitrogen no plant can photosynthesise.

(D) Only plant B turns blue-black, because nitrogen shortage forces faster starch-making.

18. In a food chain rice plant → grasshopper → frog → snake, all the energy that flows through every animal originally came from:

(A) The soil minerals absorbed by the rice plant's roots.

(B) Sunlight, captured by the rice plant during photosynthesis.

(C) The bodies of the animals as they eat one another.

(D) The oxygen the animals breathe in.

19. A leaf is left attached to a healthy plant in sunlight. A clear plastic bag is tied over it and water droplets collect inside the bag, while in a parallel set-up a leaf that has been killed (its cells dead) and bagged the same way collects far fewer droplets. The best explanation is:

(A) The living leaf loses water vapour through its working stomata (transpiration); a dead leaf cannot.

(B) The droplets are rain that leaked into the living leaf's bag only.

(C) The living leaf's xylem tubes spray water directly out of the surface.

(D) The dead leaf actually makes water, which is why it has fewer droplets.

20. A gardener finds a yellow, leafless Cuscuta vine thriving on her rose bush while the rose declines. She also finds a mushroom on a dead stump nearby that harms nothing living. Pairing each with the correct mode of nutrition:

(A) Cuscuta is a parasite; the mushroom is a saprotroph.

(B) Cuscuta is a saprotroph; the mushroom is a parasite.

(C) Both Cuscuta and the mushroom are parasites.

(D) Cuscuta is a symbiont; the mushroom is an autotroph.

21. Why is boiling a green leaf in WATER (not alcohol) included as a separate step in the starch test, distinct from the alcohol step?

(A) To dissolve out the green chlorophyll, which is alcohol's job.

(B) To add starch to the leaf before testing.

(C) To soften the leaf and break its cells so that iodine and alcohol can act properly later.

(D) To release the oxygen trapped inside the leaf.

22. A desert plant has thick green stems and almost no leaves. A classmate concludes it 'cannot photosynthesise and must steal food.' Which observation would BEST refute the classmate and explain the plant's survival?

(A) The green stems test positive for starch after a day in the sun, showing the stems photosynthesise.

(B) The plant has very long roots reaching deep water, proving roots make its food.

(C) The plant has no flowers, which shows it must be a parasite.

(D) The plant grows slowly, proving it gets food from the soil.

23. A balanced sealed terrarium with a green plant survives for weeks in good light. If the SAME terrarium is then moved to a dark cupboard for two weeks, the most likely outcome is:

(A) Nothing changes, because the plant pauses all activity in the dark.

(B) Oxygen builds up, because the plant respires faster in the dark.

(C) Carbon dioxide builds up and oxygen falls, so the plant weakens and may die.

(D) The plant becomes a saprotroph and lives off the soil.

24. Air is about 78% nitrogen gas, yet a tomato plant in nitrogen-rich AIR but in nitrogen-poor SOIL can still show stunted, pale growth. The reason is that:

(A) The leaves' stomata are too small to let nitrogen gas in.

(B) Nitrogen gas in air is a different element from the nitrogen plants need.

(C) Plants cannot absorb nitrogen gas from the air directly; it must be fixed into the soil in a usable form first.

(D) Plants do not actually require nitrogen for growth.

25. A student writes four statements about photosynthesis. Exactly ONE is INCORRECT. Which?

(A) Chlorophyll captures the light energy used in photosynthesis.

(B) Oxygen is one of the raw materials taken in for photosynthesis.

(C) Glucose is the food made, and some of it is stored as starch.

(D) Carbon dioxide and water are the raw materials used up.

26. A leaf with half its area covered by black paper for a day is destarched first, then exposed to sun with the paper still on, then tested with iodine. The result is a leaf that is blue-black on one half and brown on the other. This single experiment proves that photosynthesis needs:

(A) Chlorophyll, because the two halves differ in colour.

(B) Carbon dioxide, because the paper blocked the gas.

(C) Water, because the covered half dried out.

(D) Light, because only the uncovered (lit) half made starch.

27. Rhizobium living in a legume's root nodules and the legume plant form a partnership. From the PLANT's side, what does it gain, and what does it give?

(A) It gains food (sugars) and gives the bacteria nitrogen.

(B) It gains usable nitrogen compounds and gives the bacteria food and shelter.

(C) It gains carbon dioxide and gives the bacteria oxygen.

(D) It gains water and gives the bacteria chlorophyll.

28. Glucose made in a leaf does not all stay as glucose. Most plants quickly convert the surplus into starch for storage because starch:

(A) Can be made directly from sunlight without glucose.

(B) Is the gas that the plant breathes out.

(C) Is insoluble and so does not disturb the leaf by drawing in water the way dissolved sugar would.

(D) Is the green pigment that captures light.

29. A potato tuber, a slice of bread (from wheat) and an egg white are each tested with iodine. The potato and bread turn deep blue-black; the egg white barely changes. The best conclusion is:

(A) The potato and bread are rich in starch; the egg white is not.

(B) The egg white is the richest source of starch of the three.

(C) Iodine only detects starch in plant cells that are still alive.

(D) The blue-black colour shows the potato and bread contain chlorophyll.

30. A bean seed germinates in damp cotton in a dark drawer and still pushes up a pale shoot for several days. Which statement correctly explains AND correctly predicts what happens next?

(A) It photosynthesises in the dark using stored chlorophyll, so it grows indefinitely.

(B) The damp cotton supplies it with ready-made food forever.

(C) It has become a parasite on the cotton and will keep growing.

(D) It lives on food stored in the seed; once that runs out, without light it cannot make more food and will die.

31. An experimenter wants to show that CHLOROPHYLL (not just any leaf tissue) is required for photosynthesis. Which single set-up does this MOST directly, with the fewest extra variables?

(A) Keep a whole green plant in the dark for two days, then test a leaf.

(B) Cover half a green leaf with foil, then do the starch test.

(C) Do the starch test on a variegated leaf and compare its green and white parts after a sunny day.

(D) Seal a green plant in a jar and measure the gas.

32. During a bright noon, a healthy green leaf is simultaneously taking in carbon dioxide for one process and giving out carbon dioxide for another. Which statement is the MOST accurate description of the NET result at noon?

(A) Respiration stops at noon, so the leaf only takes in CO_2.

(B) Photosynthesis far exceeds respiration, so the leaf shows a large net intake of CO_2 and net release of O_2.

(C) The two processes cancel exactly, so no net gas exchange occurs at noon.

(D) Photosynthesis stops at noon, so the leaf only gives out CO_2.

33. A child concludes from one experiment, 'Plants need carbon dioxide to make starch,' after destarching a plant, sealing one leaf in CO2-free air and another in normal air, and finding only the normal-air leaf turns blue-black. For the conclusion to be SOUND, why was destarching at the start essential?

(A) So that the leaf would turn green again before the test.

(B) So that carbon dioxide could be added to the leaf more easily.

(C) So that the iodine would stick better to the leaf surface.

(D) So that any starch found at the end was definitely made during the experiment, not left over from before.

34. Which organism makes its OWN food, yet is most often grouped with the FOOD-MAKER in a partnership rather than studied alone?

(A) The fungus in a lichen. (B) Rhizobium in a root nodule.

(C) Cuscuta on a host plant. (D) The alga in a lichen.

35. Comparing nutrition in a green plant and a cow: the most fundamental difference is that:

(A) The cow makes its own food from grass using sunlight.

(B) The plant synthesises its food from simple inorganic substances; the cow must take in ready-made organic food.

(C) The plant eats soil organisms while the cow eats grass.

(D) Both obtain ready-made food from their surroundings.

36. A farmer grows only wheat in the same field every year and the yields steadily drop. A soil scientist says one mineral is being progressively drained. To restore it WITHOUT chemicals, the best single action is:

(A) Plant a legume crop one season and plough it back, letting Rhizobium fix nitrogen.

(B) Add more wheat seed per square metre to make the soil richer.

(C) Spray the field with extra water through the dry months.

(D) Remove all weeds, since weeds are the only thing draining the mineral.

37. A leaf's broad, flat shape and its dense packing of green cells near the upper surface both serve a single overall purpose. That purpose is to:

(A) Store as much water as possible inside the leaf.

(B) Maximise the capture of sunlight for photosynthesis.

(C) Make the leaf heavy enough to resist wind.

(D) Keep insects from landing on the leaf.

38. In the sealed-jar-with-a-candle demonstration done in BRIGHT LIGHT, a green plant keeps a small flame burning longer than an identical jar with no plant. If the same plant-and-candle jar is instead kept in the DARK, the most likely result compared with the empty dark jar is:

(A) The flame burns even longer, because the plant releases oxygen in the dark.
(B) The flame burns exactly the same, because plants do nothing in the dark.
(C) The flame goes out sooner, because in the dark the plant also uses up oxygen by respiring.
(D) The flame turns green from the plant's chlorophyll.

39. Four sentences describe modes of nutrition. Which one is INCORRECT?
(A) An autotroph builds its food from inorganic raw materials using light.
(B) A parasite takes food from a living host and harms it.
(C) An insectivorous plant photosynthesises and traps insects mainly for nitrogen.
(D) A saprotroph attaches to a living host and slowly draws food from it.

40. A plant is given carbon dioxide, water, chlorophyll and warmth, but is kept in TOTAL darkness. After a sunny-length period a destarched leaf is tested with iodine. The result and reason are:
(A) Blue-black, because chlorophyll alone can make starch in the dark.
(B) Blue-black, because warmth can replace sunlight as the energy source.
(C) Red, because darkness changes how iodine reacts.
(D) No blue-black colour, because without light the energy needed for photosynthesis is missing.

41. A pond's green algae double in mass over summer. A scientist wants to know where the EXTRA mass mostly came from. Based on photosynthesis, the largest share of that new dry mass is built from:
(A) Soil minerals soaked up from the pond bed.
(B) The water of the pond turning directly into plant body.
(C) Carbon dioxide taken from the water and air (carbon for glucose).
(D) Oxygen absorbed from the surrounding water.

42. A student says, 'Xylem and phloem both carry water in the same direction, just at different speeds.' Identify what is WRONG with this statement.
(A) Nothing is wrong; both indeed carry water upward.
(B) Both carry food, not water, so the student named the wrong substance.
(C) Xylem carries food up while phloem carries water down.
(D) Phloem carries food (mostly downward/outward from leaves), not water; only xylem carries water, and upward.

43. Two pots of the same plant are grown for three weeks: pot 1 in normal soil, pot 2 in soil deliberately kept very low in nitrogen. Pot 2's leaves turn pale yellowish and growth is stunted, but on any single sunny day a green-ish leaf from pot 2 STILL turns blue-black in the starch test. The combined results show that nitrogen is needed for:

(A) Photosynthesis directly, since pot 2 made less starch each day.

(B) Healthy growth and the green colour, but it is NOT a raw material in the day's photosynthesis itself.

(C) Nothing at all, since pot 2 still made starch.

(D) Making carbon dioxide inside the leaf for photosynthesis.

44. An insectivorous plant, a green pea plant with Rhizobium nodules, and a Cuscuta vine all have something to do with NITROGEN. Which statement correctly describes all three?

(A) The insectivorous plant gets nitrogen from insects, the pea gets it via nitrogen-fixing bacteria, and Cuscuta gets its nitrogen (with everything else) from its host.

(B) All three fix nitrogen directly from the air through their leaves.

(C) All three obtain nitrogen by photosynthesis in sunlight.

(D) None of them needs nitrogen, since plants only need carbon dioxide and water.

45. A jar with a green water plant in light shows rising oxygen. The light is then moved farther and farther away, and the rate of bubbling steadily falls until, in near-darkness, bubbling stops. The single best conclusion is:

(A) The plant runs out of chlorophyll as light is removed.

(B) The plant runs out of water as the light moves away.

(C) Photosynthesis depends on light, and its rate falls as available light decreases.

(D) Carbon dioxide leaves the jar when the light is dimmed.

46. Consider this chain of dependence written by a student: snake → frog → grasshopper → green plant, with arrows meant to show 'gets its food from'. Is the chain correct as an 'is eaten by' / energy-flow diagram, and if not, how should the arrows point?

(A) As an energy-flow ('is eaten by') diagram it should run green plant → grasshopper → frog → snake.

(B) It is already correct as an energy-flow diagram.

(C) It should run frog → green plant → snake → grasshopper.

(D) Food chains have no direction, so either way is fine.

47. A leaf disc taken from a plant kept in the dark for 48 hours is boiled in water, then alcohol, then tested with iodine, and shows NO blue-black colour. A second disc from the same plant after 6 hours of sunlight does turn blue-black. The 48-hour dark step is best described as:

(A) Destarching — emptying the leaf of stored starch so a fresh test reveals only newly made starch.

(B) Killing the chlorophyll so the leaf can never make starch again.

(C) Adding starch slowly so the dark disc should have tested positive.

(D) A control that proves iodine does not work on dark leaves.

48. Which statement best explains why nearly ALL animals, including those that eat only other animals, ultimately depend on green plants?

(A) Green plants directly feed every animal by being eaten by all of them.

(B) Green plants produce most of their oxygen at night for animals.

(C) Green plants are the producers that first capture sunlight as food and release the oxygen animals breathe.

(D) Green plants give animals their minerals straight from the soil.

49. A pale Cuscuta vine is found growing not on a plant but tangled over a wire fence, and within days it withers and dies, whereas a Cuscuta on a living shrub thrives. This observation best supports the idea that Cuscuta is:

(A) A saprotroph that needs dead matter such as the wire to feed on.

(B) An autotroph that briefly lost its chlorophyll.

(C) A parasite that cannot survive without taking food from a living host.

(D) A symbiont that helps whatever it grows on.

50. A student measures that a leaf gives out water vapour fastest on a hot, dry, windy day and slowest on a cool, humid, still day. The structure most responsible for this water loss, and the process, are:

(A) The xylem, through photosynthesis. (B) The phloem, through respiration.

(C) The chlorophyll, through food-making.(D) The stomata, through transpiration.

51. In a classroom debate a student says: 'Because autotrophs make their own food, they don't need anything from their surroundings.' Which single correction is BEST?

(A) They still need raw materials from their surroundings — carbon dioxide, water and light — to make that food.

(B) Correct — autotrophs are fully independent of their surroundings.

(C) They need to eat small animals to get started each day.

(D) They actually take ready-made food from the soil like heterotrophs.

52. Three leaves from one plant are tested after a sunny day: leaf 1 normal, leaf 2 had its lower surface (where most stomata are) smeared with a thick layer of grease, leaf 3 left normal as a check. Leaf 2 turns much less blue-black than leaves 1 and 3. The best explanation is:

(A) The grease blocked the stomata, cutting off carbon dioxide so leaf 2 made less starch.

(B) The grease blocked sunlight from reaching the leaf's chlorophyll.

(C) The grease dissolved the leaf's starch before the test.

(D) The grease added oil that prevented iodine from ever reacting.

53. Why is a green plant correctly placed at the START of essentially every food chain, while a saprotroph like a mushroom is usually NOT shown as a starting point?

(A) The mushroom cannot be in a food chain at all.

(B) The mushroom makes more food than the plant but releases it slowly.

(C) The green plant produces new food from sunlight, while the saprotroph only breaks down food that already existed.

(D) The green plant eats the mushroom to begin the chain.

54. A leaf left attached to a well-watered plant in still, humid shade shows very little transpiration, yet the plant continues to photosynthesise whenever light is available. This shows that:

(A) Transpiration is necessary for photosynthesis, so both should have stopped together.

(B) Photosynthesis causes transpiration, so low transpiration means no food was made.

(C) Water loss and food-making are the same process under different names.

(D) Transpiration and photosynthesis are different processes; reducing water loss does not stop food-making.

55. Which of these correctly traces the FULL path of a carbon atom that ends up in the starch of a leaf?

(A) From soil minerals → up the xylem → into starch directly.

(B) From carbon dioxide in the air → into glucose during photosynthesis → stored as starch.

(C) From oxygen in the air → into glucose → stored as starch.

(D) From water in the soil → into chlorophyll → into starch.

56. A teacher claims a particular plant is an autotroph. Which single observation would most strongly DISPROVE that claim?

(A) Its leaves lose water vapour on a hot day.

(B) It grows taller when given more sunlight.

(C) Its parts contain no chlorophyll and it visibly draws all its food from another living plant.

(D) It releases oxygen bubbles when brightly lit underwater.

57. A sealed transparent box holds a green plant and is kept in steady bright light for many hours. The level of which gas inside the box would a careful experiment most likely find SLIGHTLY HIGHER at the end than at the start?

(A) Oxygen. (B) Carbon dioxide.

(C) Nitrogen.

(D) Water vapour, which the plant manufactures from sugar.

58. Pick the statement that is FALSE about the starch test as a tool for studying photosynthesis.

(A) Iodine turning blue-black indicates the presence of starch.

(B) Destarching the plant first lets you attribute new starch to the experiment.

(C) Boiling in alcohol removes chlorophyll so the colour change is easy to see.

(D) A positive starch test on a leaf proves the leaf also has chlorophyll, since only green parts can ever contain starch.

59. A bean plant is grown in a sealed clear chamber where the air's carbon dioxide is steadily replenished but the LIGHT is dimmed week by week. Over the weeks the plant's growth slows and finally it begins to weaken even though CO_2, water and warmth are plentiful. The single best explanation is:

(A) The extra carbon dioxide is poisoning the plant.

(B) The plant has run out of chlorophyll because CO_2 was replenished.

(C) The plant has turned into a saprotroph feeding on the chamber.

(D) With too little light the plant cannot photosynthesise enough food to sustain its growth.

60. Which single sentence most completely and correctly summarises nutrition in plants as covered in this chapter?

(A) Green plants make their own food by photosynthesis, while some plants and organisms obtain food in special ways — parasitic, insectivorous, saprotrophic or symbiotic — and soil nitrogen is replenished by nitrogen-fixing bacteria.

(B) All plants make food only at night, and none ever depend on other organisms.

(C) Plants and animals both make their food from sunlight in exactly the same way.

(D) Plants take all their food ready-made from the soil through their roots.

Solutions — Science (Class 7), Chapter 1: Nutrition in Plants — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	C	41	C	51	A
2	A	12	C	22	A	32	B	42	D	52	A
3	C	13	C	23	C	33	D	43	B	53	C
4	B	14	A	24	C	34	D	44	A	54	D
5	B	15	A	25	B	35	B	45	C	55	B
6	D	16	D	26	D	36	A	46	A	56	C
7	D	17	B	27	B	37	B	47	A	57	A
8	B	18	B	28	C	38	C	48	C	58	D
9	B	19	A	29	A	39	D	49	C	59	D
10	D	20	A	30	D	40	D	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Net gas exchange is the difference between what photosynthesis releases and what respiration consumes. A net oxygen exchange of zero means the two opposing rates are equal, not that anything has switched off — at dawn and dusk light is just strong enough for photosynthetic oxygen output to match respiratory oxygen use. The idea that both processes stop is wrong: respiration never stops, and at these moments photosynthesis is clearly running too. 'Only respiring' is wrong because then net oxygen would be negative, not zero. 'Photosynthesis completely stopped' would also give a negative net (respiration alone), not a balance, so it cannot describe a zero crossing.

2. (A) The boiling-in-water step softens the leaf and breaks cell membranes, and the alcohol step strips out chlorophyll; together these let iodine reach the starch and let the colour be seen. Disc Q, untreated, is still green and waxy so iodine cannot get in and the green masks any faint colour — its starch is simply undetected, not absent. The stem explicitly states both discs contained starch, so 'never contained starch' contradicts the premise. Neither boiling water nor alcohol can create or add starch; they only prepare the tissue. Hence the difference is about detection, not about how much starch was made.

3. (C) The only true statement that supports a night-time oxygen drop is that in darkness photosynthesis stops while respiration continues, so plants take in oxygen and release carbon dioxide. (In practice the amount is tiny, but the mechanism is real.) 'Photosynthesise faster at night' is false — there is no light, so photosynthesis stops entirely. 'No need for oxygen at any time' is false because plants respire continuously, day and night. 'Chlorophyll absorbs oxygen in darkness' is a made-up mechanism; chlorophyll captures light energy, it does not consume oxygen. Only the respiration statement is both true and relevant.

4. (B) Algae are the producers here; the tadpoles depend directly on them for food, so losing the algae most immediately starves the tadpoles and their numbers fall. The fish depend on tadpoles, so any effect on fish comes later and indirectly — the fish do not benefit, nor do they die first, because the chain runs algae → tadpoles → fish. Tadpoles are animals (heterotrophs) and cannot make their own food from sunlight, so 'switch to making their own food' is impossible. The first and most direct link to break is the producer-to-tadpole one.

5. (B) Carbon dioxide is a raw material for photosynthesis. The leaf in ordinary air has light, chlorophyll, water and carbon dioxide, so it makes starch and turns blue-black. The leaf sealed with potassium hydroxide is starved of carbon dioxide, so it cannot make starch and stays brown. 'Both turn blue-black / light alone makes starch' is wrong because light cannot substitute for a missing raw material. 'Neither turns blue-black because destarching removed the ability' confuses removing existing starch with disabling the process — destarching only empties the leaf so any new starch comes from this experiment. 'Only the CO_2-free leaf reacts' is backwards, since removing a needed gas stops, not speeds, food-making.

6. (D) A parasite takes its food from a living host and harms it; a saprotroph feeds on dead and decaying organic matter. Cuscuta is the parasite (it wraps a living host), the mushroom is the saprotroph (it grows on rotting logs). 'Cuscuta makes some of its own food' is wrong because it lacks chlorophyll and is fully dependent. The third option simply swaps the two definitions and is therefore backwards. Cuscuta does not trap insects and mushrooms do not fix nitrogen, so the last option mislabels both. Only the living-host versus dead-matter distinction correctly separates them.

7. (D) Xylem carries water up; phloem carries food (glucose/sugars) made in the leaves to the rest of the plant. Removing only the phloem leaves the water pathway intact, so the leaves still get water and keep photosynthesising, but the sugars they make cannot travel down past the ring and accumulate above it, causing a swelling. 'Wilt immediately from no water' confuses phloem with xylem — water transport is untouched. 'Grow faster' is wrong because blocking food export hampers, not helps, the plant below the ring. 'Stop making food because leaves lose water' again wrongly assumes the water route was cut. The correct outcome follows from xylem intact, phloem cut.

8. (B) Wheat (a cereal) draws nitrogen OUT of the soil; it does not add it. Legumes such as gram host Rhizobium bacteria in their root nodules, which fix nitrogen from the air into a form plants can use, and ploughing the crop back enriches the soil further. So nitrogen peaks after year 3. The wheat years all deplete nitrogen, so 'after year 1', 'after year 4' and 'after year 2' are all wrong — and growing the same crop twice in a row depletes the soil more, it does not double the nitrogen. Only the legume year raises soil nitrogen.

9. (B) Photosynthesis needs both chlorophyll (only the green parts have it) and light. Starch will form only where BOTH conditions are met: green tissue that was also exposed to light. The foil-covered green tissue lacked light, so it makes no starch. The white centre lacks chlorophyll, so it makes no starch regardless of light — 'pale tissue stores the most' is a misconception. Shading does not concentrate starch; it removes the light needed to make it, so the covered region stays brown. Only the green-and-lit region tests positive.

10. (D) In a lichen each partner gains: the alga (which has chlorophyll) makes food for both, and the fungus supplies water, minerals and protection. Mutual benefit is symbiosis, not parasitism — a parasite harms its host. So calling it parasitism is wrong because nobody is being harmed. 'The alga harms the fungus' invents harm that does not exist. 'No living organisms, only minerals' is false — both fungus and alga are alive. 'The fungus makes the food' reverses the roles; the alga, not the fungus, does the photosynthesis. Only mutual benefit (symbiosis) correctly describes it.

11. (B) A net fall in carbon dioxide means photosynthesis is using up more CO_2 than respiration releases — and that requires chlorophyll plus light. Only the green fern photosynthesises, so its jar's CO_2 drops. The mushroom is a saprotroph with no chlorophyll; it only respires, so its CO_2 rises. Cuscuta lacks chlorophyll (it is normally a parasite) and with no host it cannot make food either; it only respires, raising CO_2. The snail is an animal that only respires, raising CO_2. Hence only the green fern lowers carbon dioxide.

12. (C) The correct equation is carbon dioxide + water → (in the presence of sunlight and chlorophyll) → glucose + oxygen. The teacher's version wrongly used oxygen as a raw material and water as a product; the fix swaps them so water is taken in on the left and oxygen is released on the right. Chlorophyll is correct and is not replaced by soil minerals — minerals are not part of the equation. The direct product is glucose, not starch (starch forms later as storage), so that swap would introduce a new error. Sunlight is the energy source; replacing it with darkness would stop the reaction. Only the oxygen/water swap repairs the equation.

13. (C) Stomata are the pores through which carbon dioxide enters and oxygen and water vapour leave. Closing them on a hot, dry day reduces water loss but also chokes off the

CO_2 supply, a raw material, so photosynthesis slows. Sunlight reaches chlorophyll through the leaf surface, not through the stomata, so 'sunlight can't pass' is wrong. Closing stomata does not destroy chlorophyll — the pigment is unaffected. Root absorption of water is a separate process and is not switched off by stomatal closure. The limiting factor here is the cut-off of carbon dioxide.

14. (A) Insectivorous plants like the pitcher plant are green and photosynthesise normally for their main (carbohydrate) food; they trap insects only to obtain nitrogen, which their boggy soil lacks. So both facts fit together: green = food-making, insect-trapping = nitrogen supply. 'Gets all its food from insects' ignores the working chlorophyll. It is not a parasite, because it does not draw food from a living host plant — it digests prey for a mineral. It is not a saprotroph, because it captures live insects, not dead decaying matter, and the green leaves are a real, used adaptation, not accidental.

15. (A) In light, the plant photosynthesises and releases oxygen as a product; the bubbles appear only in light and stop in the dark, which matches photosynthesis exactly. Carbon dioxide is a raw material taken IN, not given off, during photosynthesis, so it cannot be the bubbled gas. Water vapour does not form visible underwater bubbles, and the plant is submerged in water anyway. Nitrogen is not produced or released by the plant. The light-dependent appearance of the bubbles confirms the gas is photosynthetic oxygen.

16. (D) An autotroph makes its own food using chlorophyll and light. The mango tree, the green alga, the green-stemmed cactus and photosynthetic blue-green organisms all have chlorophyll and make their own food, so they are correctly listed. Cuscuta has no chlorophyll and parasitises a host, so it is a heterotroph — that is the single wrong entry. The cactus's green stem does photosynthesise (so iii is fine). Size has nothing to do with being an autotroph, so the alga objection is false. Trees do not take energy from the soil; they trap sunlight, so the mango objection is false too.

17. (B) The single-day starch test detects starch made from that day's photosynthesis, which needs carbon dioxide, water, light and chlorophyll — nitrogen is NOT one of the raw materials in the word equation. As long as a leaf is green and lit, it can still make starch for that day, so both can turn blue-black. (Over many weeks plant B grows poorly and yellows from nitrogen shortage, but a surviving green leaf still photosynthesises.) 'Only A reacts because nitrogen is a raw material' is wrong — nitrogen is needed for proteins/growth, not for the photosynthesis equation. 'Neither reacts' overstates nitrogen's role. 'Nitrogen shortage speeds starch-making' is invented. The trap is confusing growth nutrients with photosynthesis raw materials.

18. (B) Green plants (producers) trap sunlight's energy and lock it into food; every consumer above them — grasshopper, frog, snake — ultimately runs on that same captured solar energy passed along the chain. So the original source is sunlight via the rice plant. Soil minerals are needed for the plant's growth but are not the energy source. The animals' bodies merely pass energy along; they did not originate it. Oxygen lets animals release energy from food by respiration, but oxygen itself is not an energy source. The trail leads back to sunlight.

19. (A) Transpiration is the loss of water vapour from a leaf, mainly through the stomata, which are controlled by living cells. A living leaf with functioning stomata releases water vapour that condenses inside the bag; a dead leaf's stomatal mechanism no longer works, so far less water escapes. Rain is ruled out because both bags are sealed and treated identically. Xylem carries water up inside the leaf but does not spray it from the surface; water leaves as vapour through stomata. A dead leaf does not 'make water' — that is not a thing leaves do. The difference is living, working stomata.

20. (A) Cuscuta draws food from the living rose and harms it — that is parasitism. The mushroom lives on a dead stump, decomposing dead matter without harming anything living — that is saprotrophic nutrition. Swapping the two labels is wrong because the mushroom is on dead, not living, tissue and Cuscuta is on a living host. Calling both parasites is wrong since the mushroom harms no living host. 'Symbiont' is wrong for Cuscuta because the rose is harmed, not helped, and 'autotroph' is wrong for the mushroom because it has no chlorophyll. The living-host (parasite) versus dead-matter (saprotroph) distinction is decisive.

21. (C) The water-boiling step softens the leaf tissue and breaks open the cells, making them permeable so that the later alcohol step can reach and dissolve the chlorophyll and so iodine can later reach the starch. Removing the green colour is the ALCOHOL step's job, not the water step's, so that option swaps the two roles. Boiling adds no starch — the leaf already contains whatever it made. The step is not about releasing oxygen. Distinguishing the softening role of water from the de-greening role of alcohol is the key point.

22. (A) If the green stems turn blue-black with iodine after a sunny day, they contain freshly made starch, proving the stems themselves photosynthesise — directly refuting the 'cannot photosynthesise / must steal' claim. Long roots reaching water explain water supply, not food-making; roots do not manufacture food. Absence of flowers has nothing to do with parasitism. Slow growth does not show soil-feeding, since plants never take food (energy) from soil. The decisive evidence is the positive starch test in the green stems.

23. (C) In the light the terrarium balanced because photosynthesis used the CO_2 from respiration and respiration used the O_2 from photosynthesis. In the dark, photosynthesis stops but respiration continues, so oxygen is steadily consumed and carbon dioxide accumulates, weakening the plant. 'Nothing changes / pauses all activity' is wrong because respiration never stops. 'Oxygen builds up' is backwards — respiration removes oxygen and releases CO_2. A green plant cannot turn into a soil-feeding saprotroph. The loss of the photosynthetic half of the balance is what dooms the system.

24. (C) Although the air is rich in nitrogen gas, plant roots cannot take up that gaseous nitrogen directly; it must first be 'fixed' (by bacteria such as Rhizobium, or by lightning, or supplied as fertiliser) into a soil-borne form the roots can absorb. So nitrogen-poor soil starves the plant despite the air. The stomatal-size idea is wrong — stomata exchange CO_2, O_2 and water vapour, not the plant's mineral nitrogen supply, which comes through roots. Nitrogen in air is the same element, just in an unusable gaseous form. And plants definitely need nitrogen (for proteins and growth), which is why deficiency makes them pale and stunted.

25. (B) Oxygen is a PRODUCT released by photosynthesis, not a raw material taken in, so the statement calling it a raw material is the incorrect one. The other three are all correct: chlorophyll captures the light energy; glucose is the food made, with surplus stored as starch; and carbon dioxide and water are indeed the raw materials consumed. Spotting that oxygen is on the output side, not the input side, identifies the false statement.

26. (D) The only variable that differs between the two halves is light: both halves are green (same chlorophyll), both have access to the same air (CO_2) and the same water supply through the plant. Only the covered half was deprived of light, and only it failed to make starch — so the experiment isolates LIGHT as the necessary factor. It does not test chlorophyll, since both halves are equally green. Black paper does not seal the leaf from air, so it does not specifically deny carbon dioxide. Covering with paper does not dry the leaf of water. The controlled variable is light alone.

27. (B) In this symbiosis the bacteria fix atmospheric nitrogen into compounds the plant can use, so the plant GAINS nitrogen; in return the plant (which makes its own food by photosynthesis) supplies the bacteria with food and a sheltered home in the nodules. The first distractor reverses the exchange — the plant supplies food, not gains it from the bacteria; the bacteria supply nitrogen, not gain it. The plant gets carbon dioxide from air for photosynthesis, not from the bacteria, so the CO_2/O_2 swap is wrong. The plant does not give away chlorophyll. The correct exchange is nitrogen-in, food-and-shelter-out from the plant's view.

28. (C) Starch is an insoluble storage form; storing surplus food as insoluble starch keeps it from dissolving and disturbing the leaf's water balance, and it stores compactly. So the leaf converts glucose to starch. Starch is NOT made directly from sunlight skipping glucose — glucose is the first food product and starch comes from it. Starch is a solid carbohydrate, not a gas, so it is not what the plant breathes out (that is carbon dioxide from respiration). Starch is not a pigment — chlorophyll is the green light-capturing pigment. Only the insoluble-storage reasoning is correct.

29. (A) Iodine turns blue-black in the presence of starch. The potato and wheat-based bread are starchy stores and react strongly, while egg white (mainly protein) lacks starch and barely changes — so the egg white is NOT a starch source. 'Egg white is the richest in starch' contradicts its weak reaction. Iodine detects starch whether cells are alive or not (a potato slice and bread are not 'living' in the leaf sense), so that claim is false. The blue-black colour signals starch, not chlorophyll; chlorophyll is the unrelated green pigment. The result simply maps colour change to starch content.

30. (D) A germinating seed draws on its own stored food (in the cotyledons/endosperm), which is why it can grow a shoot even in darkness for a while. But this store is finite; without light the seedling cannot photosynthesise to replace it, so once the reserve is exhausted it will weaken and die. Seeds do not photosynthesise in the dark — there is no light, and the pale shoot shows little working chlorophyll, so 'grows indefinitely' is false. Damp cotton supplies water, not food, so it cannot feed the plant forever. Cotton is dead fibre, not a living host, so the seedling is not a parasite. The store-then-fail prediction is the correct one.

31. (C) A variegated leaf carries green (chlorophyll-containing) and white (chlorophyll-free) regions side by side under identical light, air and water; testing for starch shows it only in the green parts, isolating chlorophyll as the needed factor with no other variable changed. Keeping a plant in the dark tests LIGHT, not chlorophyll. Covering half a leaf with foil also tests LIGHT (both halves are green). Sealing a plant in a jar tests gas exchange, not chlorophyll. Only the variegated-leaf comparison varies chlorophyll alone while holding the rest constant.

32. (B) At bright noon both photosynthesis and respiration run, but photosynthesis is far faster, so the net effect is a large intake of carbon dioxide and release of oxygen. Respiration does NOT stop at noon — it runs continuously day and night — so 'respiration stops' is wrong even though its effect is masked. The two do not cancel at noon; cancellation happens only at low light (dawn/dusk). Photosynthesis certainly does not stop at bright noon — it is at its strongest. The correct picture is photosynthesis dominating with a net CO_2 intake.

33. (D) Destarching (keeping the plant in the dark so it uses up stored starch) ensures the leaves start empty; then any starch detected afterwards must have been made during the experiment, making the comparison fair and the conclusion sound. Destarching is not about turning the leaf greener — colour change is handled by the alcohol step later. It does not help add carbon dioxide; the CO_2 manipulation is done with the sealed flasks. It is unrelated to how iodine sticks. The whole point is a clean starting baseline so the result can be trusted.

34. (D) The alga in a lichen contains chlorophyll and makes food by photosynthesis (it is the food-maker of the partnership), even though it is studied as part of the lichen partnership rather than alone. The fungus in the lichen does NOT make food; it supplies water, minerals and shelter. Rhizobium fixes nitrogen but does not photosynthesise — it is not a food-maker. Cuscuta lacks chlorophyll and steals food, so it makes no food of its own. Only the lichen's alga is a self-feeding food-maker within a partnership.

35. (B) A green plant is an autotroph: it builds its own food from simple inorganic raw materials (carbon dioxide and water) using sunlight. A cow is a heterotroph: it cannot synthesise food and must eat ready-made organic food (grass). 'The cow makes its own food using sunlight' is false — animals lack chlorophyll and cannot photosynthesise. The plant does not 'eat soil organisms'; it makes food in its leaves. 'Both obtain ready-made food' is wrong because the plant makes, rather than obtains, its food. The autotroph-versus-heterotroph contrast is the fundamental one.

36. (A) Continuous wheat drains soil nitrogen. Growing a legume (with Rhizobium in its root nodules) and ploughing it back fixes atmospheric nitrogen into the soil naturally, restoring fertility without chemical fertiliser. Sowing more wheat removes even more nitrogen — it does not add any. Extra water does not supply nitrogen. Weeds compete but are not the cause of the nitrogen loss; the wheat crop itself is depleting it. The legume rotation is the natural nitrogen-restoring choice.

37. (B) A broad, flat blade presents a large surface to the sun, and the chlorophyll-rich cells just under the upper surface are positioned to receive that light — both features are about capturing maximum sunlight for photosynthesis. A thin flat leaf is poor for storing water (water-storing leaves are thick and fleshy), so water storage is wrong. Broad thin leaves are light, not heavy, and a flat shape catches wind rather than resisting it. The shape does not deter insects. Light capture ties both features together.

38. (C) In bright light the plant photosynthesises and adds oxygen, helping the flame. In the dark, photosynthesis stops but respiration continues, so the plant CONSUMES oxygen alongside the flame, using it up faster — the flame goes out sooner than even an empty jar. 'Burns even longer / releases oxygen in the dark' is wrong because there is no photosynthesis without light. 'Does nothing in the dark' ignores respiration. The flame does not turn green — chlorophyll does not colour a flame. The plant competing for oxygen in the dark is the key insight.

39. (D) A saprotroph feeds on DEAD and decaying matter, not on a living host — drawing food from a living host describes a parasite, so the saprotroph statement is the incorrect one. The autotroph statement is correct (builds food from inorganic raw materials using light). The parasite statement is correct (living host, harms it). The insectivorous-plant statement is correct (green, photosynthesises, traps insects for nitrogen). Confusing 'dead matter' with 'living host' is the trap that flags the wrong sentence.

40. (D) Photosynthesis needs light as its energy source; with carbon dioxide, water and chlorophyll present but NO light, no food is made, so the destarched leaf stays brown after the iodine test. Chlorophyll captures light but cannot make starch without light to capture, so 'chlorophyll alone in the dark' is wrong. Warmth is not a substitute for light energy in photosynthesis, so that option fails. Iodine still gives blue-black with starch or no colour without it; darkness does not turn it red. The missing light is what prevents starch formation.

41. (C) Photosynthesis builds glucose (and from it most of the plant's structural material) from carbon dioxide and water, with the carbon backbone coming from CO_2. So the bulk of new dry mass traces to carbon dioxide. Soil minerals are needed in small amounts for healthy growth but are not the main building material. Water contributes hydrogen and oxygen but the framing 'turning directly into plant body' misses that carbon from CO_2 forms the carbohydrate skeleton. Oxygen is a product released, not absorbed to build mass. The dominant raw material for new mass is carbon dioxide.

42. (D) Xylem carries water (and dissolved minerals) UPWARD from roots to leaves; phloem carries the FOOD made in the leaves to the rest of the plant. So the student is wrong in two ways: phloem carries food not water, and they do not move the same substance in the same direction. 'Nothing is wrong' endorses the error. 'Both carry food' is also wrong because xylem carries water. The last option reverses the roles (xylem-food, phloem-water), which is backwards. The correct correction is water-up in xylem, food-out in phloem.

43. (B) Pot 2 shows that lacking nitrogen causes pale, stunted growth — so nitrogen matters for building the plant (proteins, healthy green leaves). But a surviving green leaf still passes the daily starch test, showing photosynthesis (CO_2 + water + light + chlorophyll) does not directly need soil nitrogen as a raw material. So nitrogen aids growth and greenness but is not in the photosynthesis equation. 'Photosynthesis directly' is wrong because the leaf still made starch. 'Nothing at all' ignores the obvious stunting and yellowing. Leaves do not 'make carbon dioxide for photosynthesis' — CO_2 comes from the air. The nuanced split (growth yes, photosynthesis raw material no) is correct.

44. (A) Each obtains nitrogen by a different route: the insectivorous plant traps and digests insects for nitrogen; the pea hosts Rhizobium that fixes atmospheric nitrogen into the soil for it; Cuscuta, being a parasite, draws nitrogen (and all its food) from its living host. 'All fix nitrogen through their leaves' is false — leaves do not fix nitrogen. 'Obtain nitrogen by photosynthesis' confuses nitrogen supply with food-making; photosynthesis makes carbohydrate, not nitrogen compounds. 'None needs nitrogen' is false — all living plants need nitrogen for proteins. Only the three-different-routes description is correct.

45. (C) The bubbling (oxygen release) tracks the light level — strong in bright light, fading as light is reduced, stopping in darkness. This shows photosynthesis depends on light and its rate rises and falls with light intensity. Moving the light does not destroy or use up chlorophyll, so 'runs out of chlorophyll' is wrong. The plant is sitting in water, so it does not run out of water when the light moves. Dimming the light does not push carbon dioxide out of the jar. The clean link is light intensity controlling the photosynthesis rate.

46. (A) Energy flows from the producer upward: the green plant is eaten by the grasshopper, the grasshopper by the frog, the frog by the snake — so the energy-flow arrows run green plant → grasshopper → frog → snake. The student's chain points the wrong way for energy flow (it shows 'gets food from', the reverse direction). It is therefore not already correct as an energy-flow diagram. The frog-first ordering is jumbled and biologically wrong. Food chains DO have a definite direction (producer to top consumer), so 'either way is fine' is false.

47. (A) Keeping a plant in the dark for a day or two makes it use up its stored starch (destarching); the dark disc then shows no starch, confirming the baseline is empty, while the sunlit disc shows freshly made starch. The dark step does not kill chlorophyll — the leaf can photosynthesise again once lit, as the second disc proves. It does not add starch; darkness depletes starch, which is why the dark disc is negative. It is not a test of whether iodine works on dark leaves; iodine works fine — there is simply no starch to react with. Destarching is the correct description.

48. (C) Green plants are the producers: they trap sunlight's energy as food that is passed along every food chain (so even a pure meat-eater's prey traces back to plants), and they release the oxygen animals need to respire. Hence all animals ultimately depend on them. 'Directly eaten by all of them' is false — carnivores do not eat plants directly; the dependence is through the food chain. Plants release oxygen during the DAY (in light), not at night, so the night claim is wrong. Plants do not hand minerals from the soil to animals; the universal dependence is about food energy and oxygen.

49. (C) Cuscuta lacks chlorophyll and cannot make its own food; on a living shrub it taps the host and thrives, but on a lifeless wire it has no host to draw food from and dies. This is exactly parasitic behaviour — dependence on a living host. A saprotroph feeds on dead organic matter, but a metal wire is neither organic nor something Cuscuta can decompose, so that fails. It is not an autotroph — it never makes its own food, chlorophyll or not. It is not a symbiont, because it harms its host rather than helping it. The death on a non-living support pinpoints parasitism.

50. (D) Water vapour escapes from the leaf mainly through the stomata, and the process of losing this water vapour is called transpiration; it speeds up in hot, dry, windy conditions and slows in cool, humid, still air. Xylem carries water inside the plant but the loss to the air is through stomata, and photosynthesis is food-making, not water loss. Phloem carries food, and respiration is energy release, neither of which is the water-vapour-loss process. Chlorophyll captures light for food-making and is unrelated to releasing water vapour. Stomata + transpiration is the correct pairing.

51. (A) Autotrophs make their own food, but they are not independent of their surroundings: they draw carbon dioxide from the air, water (and minerals) from the soil, and light energy from the sun to carry out photosynthesis. So the claim is wrong. They are not 'fully independent'. They do not need to eat small animals (that would describe insectivorous plants, and even those photosynthesise). They do not take ready-made food from the soil — that misconception is exactly what autotrophy disproves. The correction is that autotrophs still depend on raw materials from their environment.

52. (A) Most stomata are on the lower surface; greasing it seals these pores, blocking carbon dioxide entry. With its CO_2 supply cut, leaf 2 makes less starch and tests less blue-black, while the unsmeared leaves 1 and 3 test strongly. The grease is on the LOWER surface, while light falls on the upper surface, so it did not block sunlight to the chlorophyll. Grease does not dissolve starch already inside the leaf cells. The iodine test still works after the leaf is boiled in water and alcohol (which would remove surface grease), so 'oil prevents iodine reacting' is not the reason — the real cause is reduced CO_2 intake.

53. (C) A green plant is a producer — it captures sunlight to make brand-new food, so energy enters the living world through it, placing it at the start of food chains. A saprotroph such as a mushroom does not make new food; it decomposes already-existing dead matter, so it is not an originating producer. The mushroom can appear in food webs (as a decomposer), so 'cannot be in a food chain at all' is too strong. It does not make food, let alone more than the plant. The green plant does not eat the mushroom — plants make their own food and do not eat fungi. The make-new-food versus break-down distinction is the reason.

54. (D) Photosynthesis (making food from CO_2, water and light) and transpiration (loss of water vapour from the leaf) are distinct processes. In humid, still shade transpiration is low, yet whenever light is present the leaf still photosynthesises — showing one does not control the other in lock-step. 'Transpiration is necessary for photosynthesis' is wrong, since food-making continued despite low water loss. 'Photosynthesis causes transpiration / low transpiration means no food' wrongly ties the two together. They are not the same process under different names — one builds food, the other releases water vapour. The observation proves they are independent.

55. (B) The carbon in a leaf's starch comes from atmospheric carbon dioxide: photosynthesis uses CO_2 (and water) to build glucose, and surplus glucose is converted to starch for storage. So the carbon atom travels CO_2 → glucose → starch. Soil minerals supply other nutrients but not the carbon skeleton of starch, and water travels up the xylem, not carbon-bearing minerals. Oxygen is a product released, not the source of the carbon in glucose. Water provides hydrogen and oxygen, and chlorophyll is a pigment, not a carbon store leading to starch. The carbon traces back to carbon dioxide.

56. (C) An autotroph makes its own food using chlorophyll and light. If a plant has NO chlorophyll and takes all its food from another living plant, it cannot be making its own food — that directly disproves autotrophy (it is a parasite). Losing water vapour on a hot day (transpiration) happens in ordinary autotrophs too, so it proves nothing. Growing taller in more sunlight is exactly what an autotroph does (more light, more food), so it supports rather than disproves the claim. Releasing oxygen bubbles when lit is a sign of photosynthesis, again consistent with being an autotroph. Only the no-chlorophyll, host-fed observation disproves it.

57. (A) In steady bright light the plant photosynthesises faster than it respires, so on balance it takes in carbon dioxide and releases oxygen — oxygen ends up slightly higher. Carbon dioxide would be slightly LOWER, since it is consumed faster than respiration releases it. Nitrogen is not part of photosynthesis or respiration, so its level barely changes — it is not the gas that rises due to the plant's activity. Water vapour may rise from transpiration, but the option's reason ('manufactures from sugar') is false — the plant does not make water from sugar. The clearest, correctly-reasoned rise is in oxygen.

58. (D) A positive starch test only shows starch is present; it does not by itself prove the tissue has chlorophyll, because starch can be stored in non-green parts too (for example potato tubers and wheat grains test positive yet are not green). So that statement is false. The other three are true: iodine turns blue-black with starch; destarching first makes new starch attributable to the experiment; and boiling in alcohol strips chlorophyll so the iodine colour shows clearly. The false claim is the one over-reading a positive result as proof of chlorophyll.

59. (D) Light is the energy source for photosynthesis. As light is dimmed, the plant makes less and less food despite abundant raw materials, so growth slows and the plant weakens — light has become the limiting factor. Replenished carbon dioxide is helpful, not poisonous, so 'CO_2 poisoning' is wrong. Replenishing CO_2 does not deplete chlorophyll; the two are unrelated. A green bean plant cannot turn into a saprotroph and there is no dead matter in a clean chamber to feed on. Insufficient light limiting photosynthesis is the correct reason.

60. (A) The chapter's central idea is that green plants are autotrophs making food by photosynthesis, while certain plants and organisms feed in special ways (parasites like Cuscuta, insectivorous plants, saprotrophs like mushrooms, symbionts like lichen and Rhizobium), and soil nitrogen is restored by nitrogen-fixing bacteria — the correct option captures all of this. 'Make food only at night' is false (photosynthesis needs light, by day). 'Plants and animals both make food the same way' is false — animals are heterotrophs and cannot photosynthesise. 'Take all food ready-made from the soil' is the classic misconception the whole chapter overturns. Only the comprehensive option is correct.